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Mid – 2 Module wise important questions.

MODULE 3: GAN Evaluation and Training Objectives

1. Explain the Inception Score (IS) with its formula. How do image quality and diversity affect the score?

2. Explain WGAN-GP and the purpose of the gradient penalty.

3. The feature means of real and generated images are (2,3) and (4,3), respectively. Both distributions have identity covariance matrices. Calculate FID using the standard formula and interpret the result.

4. Explain the difference between a GAN discriminator and a WGAN critic.

5. Explain the objective functions of the generator and discriminator in Min-Max GAN.

6. What is mode collapse in GANs? Explain its effect on generated-image diversity and evaluation.

7. Explain the role of a pretrained Inception network in calculating IS and FID.

8. For a real sample, $D(x) = 0.8$, and for a generated sample, $D(G(z)) = 0.25$. Calculate the discriminator log-likelihood contribution, using natural logarithms.

9. Explain the training objectives of the original Min-Max GAN and the non-saturating GAN. Why is the non-saturating objective often preferred?

10. Describe the Lipschitz constraint in WGAN and explain why it is necessary.

11. Explain Frechet Inception Distance (FID), including the role of feature means and covariance matrices.

12. Explain the architecture and working principle of Wasserstein GAN (WGAN). What is the role of the critic and the Lipschitz constraint?

13. Discuss practical signs that may indicate a GAN is ready to stop training.

14. Explain the non-saturating generator loss and explain how it can improve generator learning.

15. A WGAN critic gives real and generated samples average scores of 0.9 and 0.4. Calculate the Wasserstein gap.

16. Explain how increasing generator and discriminator network depth affects feature learning, gradient flow and GAN training stability.

17. Explain the Inception Score (IS) and describe one limitation of this evaluation metric.

18. Explain why IS and FID should be interpreted carefully rather than used as the only measures of GAN quality.

19. Why can increasing the depth of a GAN generator or discriminator make training more difficult? Discuss gradient flow and stability.

20. Explain how the quality and diversity of generated images can affect the evaluation of a GAN.

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